Real-Time Data Engineering Project Status Report

This report summarizes the current progress, completed modules, and upcoming tasks for the real-time Twitter sentiment data engineering pipeline project developed using Kafka streaming architecture.

Completed Components

Component	Status	Description
Dataset Collection	Completed	Collected Twitter sentiment dataset from Kaggle.
CSV Processing	Completed	Prepared and cleaned CSV dataset.
JSON Conversion	Completed	Converted CSV data into JSON format for streaming.
Kafka Producer	Completed	Implemented Kafka producer to stream tweets.
Kafka Consumer	Completed	Implemented Kafka consumer to receive messages.
Real-Time Streaming	Completed	Verified producer-consumer communication.

Project Completion Status

The project has currently reached approximately **60% completion**. The real-time streaming architecture using Apache Kafka has been successfully implemented. The system can now stream JSON tweet data from the producer to the consumer in real time.

Upcoming Tasks

Next Module	Purpose
MongoDB Integration	Store streamed tweet data into a NoSQL database.
PySpark Processing	Perform real-time analytics and transformations.
Dashboard Development	Visualize streaming sentiment insights.
Deployment	Containerize and deploy pipeline using Docker or cloud services.

Current Pipeline Architecture

Kaggle CSV Dataset → JSON Conversion → Kafka Producer → Kafka Topic → Kafka Consumer

Conclusion

The project has successfully completed the ingestion and streaming stages of the data engineering pipeline. The next phase will focus on database storage, distributed processing, and real-time analytics visualization.